

Geometry of Splats

Definition (Splat Measures, Splat Functions)

We call $\mathcal{S}_{p,d} := \mathcal{P}(\mathbb{R}^p \times \text{BW}_\rho(\mathbb{R}^d))$ the family of *splat measures*. A function f is a *splat function* if there exists $\mu \in \mathcal{S}_{p,d}$ so that:

$$f(x) = f_\mu(x) := \mathbb{E}_{v, \bar{\rho} \sim \mu}[v \bar{\rho}(x)]$$

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Classical WFR theory: minimize $\mathcal{F} : \mathcal{P}_{2,ac}(\mathbb{R}^d) \rightarrow \mathbb{R}$ such as,

- Potential energy: $\mathcal{F}(\mu) = \int V(x) \mu(dx)$
- Internal energy: $\mathcal{F}(\mu) = \int U(\mu(x)) dx$
- Interaction energy: $\mathcal{F}(\mu) = \int K(x, y) \mu(dx) \mu(dy)$
- Ex. divergences: $\mathcal{F}(\mu) = \text{KL}(\mu \mid \pi), \chi^2(\mu \mid \pi), \dots$

Splats: minimize $\mathcal{R}(\mu) := \mathbb{E}_{x,y} \left[\|\mathcal{A}[f_\mu](x) - y\|^2 \right]$